GUJARAT TECHNOLOGICAL UNIVERSITY
(Established under Gujarat Act No. 20 of 2007)
ગુજરાતટેકનોલોજીકલ યુનિવર્સિટી
(ગુજરાત અધિનિયમ ક્રમાંકઃ ૨૦/૨૦૦૭ દ્વારા સ્થાપિત)

Annexure 1
Enrollment no:
**220170107129**

## STUDENT'S WEEKLY RECORD OF INTERNSHIP

NAME OF STUDENT: **CHETAN KANAIYALAL SHARMA**

DIARY OF THE WEEK: Dt: 12-01-2026 TO: 16-01-2026

DEPARTMENT: COMPUTER ENGINEERING SEM: 8

NAME OF THE ORGANISATION: BACANCY SERVICES PRIVATE LIMITED

NAME OF THE PLANT/SECTION/DEPARTMENT: AI/ML

NAME OF OFFICER INCHARGE OF THE PLANT/SECTION/DEPARTMENT: MEHUL BUDASNA

DESCRIPTION OF THE WORK DONE IN BRIEF

**CSS:**

- Applied CSS selectors and core styling techniques for structured UI design.
- Understood and implemented the CSS box model (margin, border, padding, content).
- Built responsive layouts using Flexbox.
- Designed grid-based page structures using CSS Grid.
- Practiced spacing, positioning, and display properties for layout control.

**Descriptive Statistics:**

- Studied fundamentals of statistics, including population vs sample and data types.
- Analyzed measures of central tendency (mean, median, mode).
- Evaluated measures of dispersion (range, variance, standard deviation, coefficient of variation).
- Created frequency distribution tables and univariate visualizations.
- Performed bivariate analysis: categorical–categorical, numerical–numerical, and categorical–numerical.
- Explored quantiles, percentiles, and five-number summary.
- Visualized data using boxplots and scatterplots.
- Computed and interpreted covariance and correlation; examined correlation vs causation.
- Practiced multi-variable visualization techniques.

**Probability & Distributions:**

- Studied random variables and foundational probability concepts.
- Explored probability distributions and their classifications.
- Implemented PMF, PDF, and CDF with conceptual understanding.
- Analyzed cumulative distribution functions for discrete and continuous cases.
- Learned density estimation techniques (parametric and non-parametric).
- Applied Kernel Density Estimation (KDE) for distribution approximation.

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STUDENT'S WEEKLY RECORD OF INTERNSHIP

NAME OF STUDENT: **CHETAN KANAIYALAL SHARMA**

DIARY OF THE WEEK: Dt: 19-01-2026 TO: 23-01-2026

DEPARTMENT: COMPUTER ENGINEERING SEM: 8

NAME OF THE ORGANISATION: BACANCY SERVICES PRIVATE LIMITED

NAME OF THE PLANT/SECTION/DEPARTMENT: AI/ML

NAME OF OFFICER INCHARGE OF THE PLANT/SECTION/DEPARTMENT: MEHUL BUDASNA

DESCRIPTION OF THE WORK DONE IN BRIEF

**Python & Statistical Foundations:**

**Python:**

- Began Core Python training covering syntax and execution flow.
- Understood variables, object references, and memory behaviour.
- Implemented reusable functions and explored lambda expressions.
- Applied list, dictionary, and set comprehensions for concise data handling.

**Probability & Distributions:**

- Studied Normal Distribution: equation, parameters, properties, skewness, and CDF.
- Applied standard normal transformation (z-score), empirical rule, and z-table usage.
- Explored practical uses of normal distribution in data science and 2D density visualization.
- Analyzed non-Gaussian distributions: Uniform, Log-normal, Pareto, Bernoulli, and Binomial (with formulas and criteria).
- Examined kurtosis, excess kurtosis, and distribution comparison using QQ plots.

**Data Transformations:**

- Performed mathematical transformations for normalization and variance stabilization.
- Applied log, reciprocal, square-root, power, Box-Cox, and Yeo-Johnson transformations.

**Sampling & Inference:**

- Studied Central Limit Theorem (intuition, assumptions, proof overview, and coding implementation).
- Understood sampling distributions and their role in statistical inference.
- Learned population vs sample, parameter vs estimate, and point estimation concepts.
- Constructed and interpreted confidence intervals using z- and t-procedures.
- Implemented confidence interval calculations programmatically and analyzed real case studies.

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## STUDENT'S WEEKLY RECORD OF INTERNSHIP

NAME OF STUDENT: **CHETAN KANAIYALAL SHARMA**

DIARY OF THE WEEK: Dt: 26-01-2026 TO: 30-01-2026

DEPARTMENT: COMPUTER ENGINEERING SEM: 8

NAME OF THE ORGANISATION: BACANCY SERVICES PRIVATE LIMITED

NAME OF THE PLANT/SECTION/DEPARTMENT: AI/ML

NAME OF OFFICER INCHARGE OF THE PLANT/SECTION/DEPARTMENT: MEHUL BUDASNA

DESCRIPTION OF THE WORK DONE IN BRIEF

**Python (Advanced Concepts):**

- Studied Object-Oriented Programming principles in Python.
- Implemented classes, constructors, class methods, and static methods.
- Explored dunder (magic) methods and object behavior customization.
- Practiced deep vs shallow copy and memory implications.
- Utilized *args and **kwargs for flexible function design.
- Learned multiprocessing and multithreading fundamentals for parallel execution.

**Statistical Inference & Hypothesis Testing:**

- Understood the framework of hypothesis testing: null/alternate hypotheses, significance level, rejection region, and error types (Type I & II).
- Performed z-tests and t-tests; interpreted p-values in statistical decision-making.
- Implemented one-tailed and two-tailed tests in practical scenarios.
- Applied single-sample, independent two-sample, and paired t-tests with coding examples.
- Studied Chi-square distribution and conducted goodness-of-fit and independence tests.
- Explored ANOVA and F-distribution, including assumptions, geometric intuition, and Python implementation.
- Analyzed post-hoc testing and limitations of repeated t-tests for multiple categories.
- Examined applications of hypothesis testing techniques in machine learning workflows.

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STUDENT'S WEEKLY RECORD OF INTERNSHIP

NAME OF STUDENT: **CHETAN KANAIYALAL SHARMA**

DIARY OF THE WEEK: Dt: 02-02-2026 TO: 06-02-2026

DEPARTMENT: COMPUTER ENGINEERING SEM: 8

NAME OF THE ORGANISATION: BACANCY SERVICES PRIVATE LIMITED

NAME OF THE PLANT/SECTION/DEPARTMENT: AI/ML

NAME OF OFFICER INCHARGE OF THE PLANT/SECTION/DEPARTMENT: MEHUL BUDASNA

DESCRIPTION OF THE WORK DONE IN BRIEF

**SQL:**

- Completed SQL fundamentals training.
- Executed core queries: SELECT, INSERT, UPDATE, DELETE.
- Implemented JOIN operations for relational data retrieval.
- Studied basic query optimization techniques.

**Linear Algebra for Machine Learning:**

- Understood tensors (0D to nD), including rank, axes, and shape with practical examples.
- Studied vectors: representation, row vs column form, Euclidean distance, scaling, shifting, and vector arithmetic.
- Computed dot product, vector angles, and formulated hyperplanes.
- Learned matrix fundamentals: types, operations (addition, multiplication, transpose), and scalar transformations.
- Computed determinant, minor, cofactor, adjoint, and inverse of matrices.
- Solved systems of linear equations using matrix methods.
- Explored geometric intuition: basis vectors, linear transformations (2D/3D), and matrix multiplication as transformation composition.
- Analyzed determinant significance, non-singularity, and conditions for matrix invertibility.
- Examined limitations of non-square matrices and non-commutativity in matrix multiplication.

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STUDENT'S WEEKLY RECORD OF INTERNSHIP

NAME OF STUDENT: **CHETAN KANAIYALAL SHARMA**

DIARY OF THE WEEK: Dt: 09-02-2026 TO: 13-02-2026

DEPARTMENT: COMPUTER ENGINEERING SEM: 8

NAME OF THE ORGANISATION: BACANCY SERVICES PRIVATE LIMITED

NAME OF THE PLANT/SECTION/DEPARTMENT: AI/ML

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DESCRIPTION OF THE WORK DONE IN BRIEF

**FastAPI (Python):**

- Completed FastAPI fundamentals including project structuring and routing.
- Designed APIs using Pydantic models for validation and serialization.
- Implemented query/path parameters, middleware, and dependency injection.
- Integrated JWT-based authentication and secure password hashing.
- Developed background tasks for asynchronous processing.

**Machine Learning Foundations:**

- Studied ML fundamentals: history, types (supervised, unsupervised, semi-supervised, reinforcement), and learning paradigms (batch vs online).
- Compared instance-based and model-based learning approaches.
- Analyzed real-world ML challenges: data quality, overfitting/underfitting, cost, and feature relevance.
- Understood ML development lifecycle and project planning strategies.

**Linear & Multiple Regression:**

- Implemented Simple Linear Regression (theory, intuition, and from-scratch coding).
- Derived slope and intercept mathematically and evaluated using MAE, MSE, RMSE, $R^2$, and Adjusted $R^2$.
- Studied Multiple Linear Regression: mathematical formulation, error function, and parameter estimation.

**Optimization & Gradient Descent:**

- Understood ML models as mathematical functions with loss optimization.
- Studied parametric vs non-parametric models and loss function selection.
- Implemented Gradient Descent from scratch with visualization and learning rate analysis.
- Explored Batch, Stochastic, and Mini-batch Gradient Descent with comparisons and use cases.

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## STUDENT'S WEEKLY RECORD OF INTERNSHIP

NAME OF STUDENT: **CHETAN KANAIYALAL SHARMA**

DIARY OF THE WEEK: Dt: 16-02-2026 TO: 20-02-2026

DEPARTMENT: COMPUTER ENGINEERING SEM: 8

NAME OF THE ORGANISATION: BACANCY SERVICES PRIVATE LIMITED

NAME OF THE PLANT/SECTION/DEPARTMENT: AI/ML

NAME OF OFFICER INCHARGE OF THE PLANT/SECTION/DEPARTMENT: MEHUL BUDASNA

DESCRIPTION OF THE WORK DONE IN BRIEF

**Classification Concepts**

- Studied classification fundamentals, advantages, and common limitations (bias, variance, class imbalance).
- Analyzed confusion matrix–based evaluation and model trade-offs.
- Understood ROC curve and AUC interpretation for threshold-independent performance measurement.

**Core Classification Algorithms**

- Implemented and compared Logistic Regression, Decision Tree, Random Forest, SVM, KNN, and Naive Bayes.
- Studied key hyperparameters and model tuning strategies.
- Analyzed underlying assumptions (linearity, independence, margin maximization, etc.).
- Evaluated practical applications across structured and real-world datasets.

**Ensemble Learning**

- Studied ensemble strategies: Voting, Bagging, Boosting, and Stacking.
- Implemented AdaBoost, Gradient Boosting, and XGBoost.
- Analyzed application scenarios and bias–variance trade-offs.
- Examined assumptions and performance benefits of ensemble methods.

**Kaggle Competitions**

- Participated in competitive ML problem-solving on real-world datasets.
- Applied end-to-end workflow: data preprocessing, model building, tuning, and evaluation.
- Gained experience in leaderboard-based validation and iterative improvement.

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STUDENT'S WEEKLY RECORD OF INTERNSHIP

NAME OF STUDENT: **CHETAN KANAIYALAL SHARMA**

DIARY OF THE WEEK: Dt: 23-02-2026 TO: 27-02-2026

DEPARTMENT: COMPUTER ENGINEERING SEM: 8

NAME OF THE ORGANISATION: BACANCY SERVICES PRIVATE LIMITED

NAME OF THE PLANT/SECTION/DEPARTMENT: AI/ML

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DESCRIPTION OF THE WORK DONE IN BRIEF

**Unsupervised Learning:**
- Studied fundamentals of unsupervised learning and its real-world applications in clustering, anomaly detection, and pattern discovery.
- Analyzed challenges such as absence of labels and evaluation complexity.

**Outliers & Anomaly Detection:**
- Identified types of outliers and applied detection/treatment techniques.
- Implemented Isolation Forest, Local Outlier Factor (LOF), and One-Class SVM for anomaly detection.
- Explored applications in fraud detection and system monitoring.

**Feature Engineering & Dimensionality Reduction:**
- Applied linear methods (PCA, LDA) and non-linear methods (Kernel PCA, t-SNE, UMAP).
- Used techniques for visualization, noise reduction, and efficiency improvement.

**Clustering Methods:**
- Explored centroid-based, hierarchical, and density-based clustering approaches.
- Implemented K-Means (variants), GMM, Agglomerative, Divisive, DBSCAN, HDBSCAN, and OPTICS.
- Analyzed assumptions and practical applications of each method.

**Clustering Evaluation:**
- Used Silhouette, Davies–Bouldin, Inertia, Calinski–Harabasz, ARI, NMI, Homogeneity, and Completeness metrics.
- Applied Elbow Method and Gap Statistic for optimal cluster selection.

**Association Rule Mining:**
- Implemented Apriori and FP-Growth algorithms.
- Studied applications in market basket analysis and recommendation systems.

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STUDENT'S WEEKLY RECORD OF INTERNSHIP

NAME OF STUDENT: **CHETAN KANAIYALAL SHARMA**

DIARY OF THE WEEK: Dt: 02-03-2026 TO: 06-03-2026

DEPARTMENT: COMPUTER ENGINEERING SEM: 8

NAME OF THE ORGANISATION: BACANCY SERVICES PRIVATE LIMITED

NAME OF THE PLANT/SECTION/DEPARTMENT: AI/ML

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DESCRIPTION OF THE WORK DONE IN BRIEF

**Deep Learning Foundations & Neural Networks**

- Studied fundamentals of Deep Learning, its history, neural network types, and comparison with Machine Learning.
- Understood Perceptron (geometric intuition, training mechanism, loss functions, and limitations) and its extension to Multi-Layer Perceptron (MLP).
- Learned forward propagation and backpropagation (conceptual understanding and mathematical formulation).

**Core Neural Network Components**

- Analyzed loss functions (Hinge, Binary Cross-Entropy) and activation functions (Sigmoid, Tanh, ReLU and variants).
- Studied vanishing/exploding gradient problems and weight initialization methods (Xavier, He).
- Explored regularization techniques: L1/L2 regularization, Dropout, Early Stopping, and Batch Normalization.

**Optimization in Deep Learning**

- Understood Gradient Descent variants (Batch, Stochastic, Mini-batch).
- Studied advanced optimizers: SGD with Momentum, NAG, AdaGrad, RMSProp, and Adam.
- Learned hyperparameter tuning concepts and optimization strategies.

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## STUDENT'S WEEKLY RECORD OF INTERNSHIP

NAME OF STUDENT: **CHETAN KANAIYALAL SHARMA**

DIARY OF THE WEEK: Dt: 05-01-2026 TO: 09-01-2026

DEPARTMENT: COMPUTER ENGINEERING SEM: 8

NAME OF THE ORGANISATION: BACANCY SERVICES PRIVATE LIMITED

NAME OF THE PLANT/SECTION/DEPARTMENT: AI/ML

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DESCRIPTION OF THE WORK DONE IN BRIEF

**HTML & Git**

- Learned and implemented standard HTML document structure and boilerplate setup.
- Explored semantic HTML elements and proper tag usage.
- Built structured content using headings, paragraphs, and lists (ordered & unordered).
- Designed and developed functional HTML forms.
- Implemented various input types, labels, and button controls.
- Applied built-in form validation using HTML attributes.
- Ensured clean markup with proper nesting and structural consistency.
- Attended introductory session on Git and GitHub fundamentals.
- Practiced core Git workflow: clone, add, commit, and push operations.